VenueTuner

System processor and measurement platform for live sound: tune any PA from one Mac and an audio interface.

User manual.

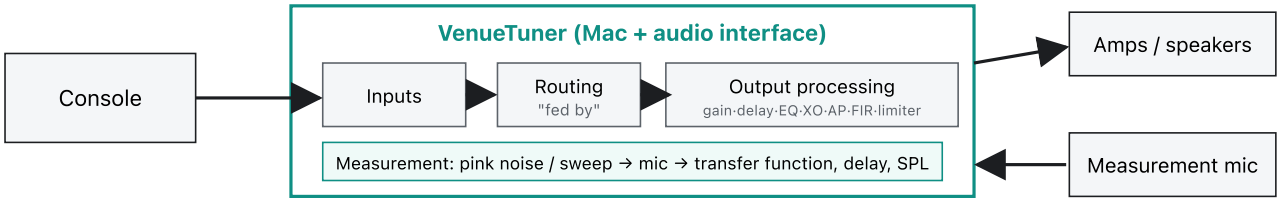
`Version 0.1 · macOS 13 Ventura or later · Apple Silicon & Intel`

Contents

1. What VenueTuner is
2. What you need
3. Installing
4. Quick start: tune a system in 15 steps
5. Settings window
6. Live window
7. Processing reference
8. Measurement window
9. Automatic tuning
10. Sound level, calibration and limits
11. Snapshots, presets and undo
12. Safety and live use
13. Keyboard shortcuts
14. Troubleshooting
15. Files, data and glossary

1 What VenueTuner is

VenueTuner turns a Mac and an audio interface into a **loudspeaker system processor** with a **measurement platform** built in. It sits between the mixing console and the amplifiers or powered speakers:



The console feeds VenueTuner; each output is processed and sent to its amplifier. A measurement mic lets you see and correct what the system and the room do.

- **Processing**: you create only the inputs and outputs you need. Each output has gain, polarity, delay, 16-band parametric EQ, high/low-pass crossovers, all-pass filters, an FIR filter and a limiter.

- **Measurement**: a dual-channel analyzer (transfer function, phase, coherence, impulse response), RTA, log sweeps, sound level in dB SPL, room reverberation (RT60).
- **Automatic tuning**: sequenced measurements over several mic positions, automatic delay alignment, sub/top phase alignment, automatic EQ. Every automatic result is written as normal, editable settings.
- **Live use**: snapshots, undo, speaker presets, air-absorption compensation, sound level against the French 2017 limits.

It works with any loudspeaker brand: you build your own speaker presets (crossover, EQ, FIR) or import the FIR files a manufacturer publishes.

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2 What you need

Item	Recommendation
Mac	macOS 13 Ventura or later, Apple Silicon or Intel. A dedicated machine is best for shows.
Audio interface	A wired USB or Thunderbolt interface (Core Audio, class compliant). Enough inputs for the console feed(s) and measurement mic(s), enough outputs for your speaker feeds. Network audio (Dante, AVB) is not supported in this version.
Measurement mic	An omnidirectional measurement microphone with 48 V phantom power, ideally with its calibration file. Optional: an acoustic calibrator (94 dB) for accurate SPL.
Cables	Console outputs → interface inputs; interface outputs → amplifiers or powered speakers. Optional: a short cable from one output to one input to calibrate the interface latency.

> **The Mac's built-in microphone and speakers**
> They work for trying the software, but Apple's processing adds about 45 ms of latency and changes the frequency response. Use them to learn the software, not to tune a system.

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3 Installing

1. Open `VenueTuner-0.1.0-universal.dmg` and drag **VenueTuner** onto **Applications**.
2. **First launch (once per version).** VenueTuner isn't distributed through the App Store, so macOS asks you to confirm:
   - **macOS 15 Sequoia and later**: double-click VenueTuner and click *Done* on the warning. Open **System Settings › Privacy & Security**, scroll down to "VenueTuner

was blocked", click **Open Anyway** and confirm.
    - **macOS 13–14**: right-click (or Ctrl-click) VenueTuner in Applications → **Open** → **Open**.
    - Alternative, in Terminal: `xattr -dr com.apple.quarantine /Applications/VenueTuner.app`

3. When macOS asks for **microphone access**, allow it. Without it the measurement inputs stay silent.

Two windows open: **Measurement** (top) and **Live** (bottom). **Settings** opens with ⌘ , . The windows remember their position.

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4 Quick start: tune a system in 15 steps

> **Before you start**
> VenueTuner always starts with **all outputs muted**, and they stay muted until you unmute them. Turn the amplifiers on last and keep them low until the patch is checked.

1. **Choose the interface**: ⌘ , › **Audio**: pick the output and input device, 48000 Hz and a buffer of 64 (or 128 if you hear dropouts), then **Apply**.
2. **Create your inputs**: Settings › **Inputs** › **+ Input** for each console feed (e.g. "Console L" on interface input 1, "Console R" on input 2), and **+ Measurement mic** for each mic. Turn on phantom power on the interface for the mic.
3. **Create your outputs**: Settings › **Outputs**: **+ Stereo pair**, **+ Sub**, **+ Output** for fills and delays. Pick each output's interface channel, and under **fed by** choose which inputs feed it.
4. **Unmute**: in the Live window (⌘ 2), check the levels, then **Unmute outputs** in the red banner.
5. **Choose the mic**: Measurement (⌘ 1) › Measure tab › **Mic**. Keep **Reference** on *Internal loopback*.
6. **Pink noise**: under Generator, light the output(s) to measure, set the level low (−40 dBFS), press **G** and bring the level up slowly.
7. **Set the mic gain**: in **Mic gain vs reference**, adjust the interface's mic preamp until the hint says ✓ *matched*.
8. **Calibrate the level** (recommended): **Calibrate…** next to the SPL value; see section 10.
9. **Look at the response**: the magnitude, phase and coherence plots fill in after a few seconds. Faded parts have low coherence and can't be trusted.

10. **Define the measurement plan**: **Matrix** tab: **One per output** creates a group per output; add mic positions (FOH, front, balcony…).
11. **Measure**: put the mic at the first position and press ▶ in its column. Each group plays alone and is captured and checked. Move the mic and repeat.
12. **Align the timing**: **Align** tab: *Auto-align* for tops, fills and delays; *Sub ↔ top phase alignment* for the crossover.
13. **EQ**: in the Matrix tab press **avg** for a group, then the **Auto-EQ** tab, **Compute** and **Apply**. Or edit by hand in the Live window's EQ editor.
14. **Check**: measure again; play music and listen.
15. **Save**: Live window › Snapshots › name › **Save**. Save speaker presets for reuse (section 11).

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5 Settings window ⌘,

Audio

Interface (output and input device), sample rate and buffer size. Smaller buffers mean less latency but more risk of dropouts (XRUN in the status bar). Changing the interface mutes all outputs.

Where the latency comes from lists the latency of each direction: device, safety offset, stream (hardware DSP), and VenueTuner's own buffer. Pro interfaces usually total 2–6 ms round trip at a buffer of 64.

Inputs

Create only the inputs you use. Each has a name, a type and an interface channel:

Type	Use
program	Sound to process (console feeds). Only program inputs can feed outputs.
measurement mic	Measurement only. **Never** sent to the speakers, even if you try to route it.
loopback	A physical reference (an output cabled back to an input), for the most exact measurements.

Outputs

One output per speaker feed: name, colour, type, interface channel and **fed by** (which inputs, at which level in dB). Presets: *Stereo pair* (L/R from the first two program inputs) and *Sub* (L+R at

−6 dB each, low-pass LR24 at 100 Hz). ↑↓ reorder, ⧉ duplicates an output with its processing, × deletes it. A channel used by another output can't be picked twice.

Calibration

Interface latency: the latency the driver reports is used automatically. For precision, cable an output to an input and press **Calibrate** (do it once per interface, sample rate and buffer). That output's processing is bypassed while it measures, so its crossovers and EQ don't end up in the interface latency. **Measurement microphone calibration file**: load the mic's .cal / .txt file to correct its frequency response.

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6 Live window ⌘2

Toolbar

Control	What it does
MUTE ALL	Silences every output immediately. Click again to unmute.
SPL reading	The main mic's level (e.g. *LAS 95.0 dB*). "≈ … uncal." means it's an estimate; click to calibrate.
LAeq 15′ / LCeq 15′	Levels over the last 15 minutes against the limits (France 2017 by default): normal, amber within 3 dB, red over. Click to change the limits.
AIR	Air-absorption compensation (section 7).
↶ ↷	Undo / redo (⌘ Z / ⇧ ⌘ Z).

Output strips

One strip per output you created: name, interface channel (and preset), output meter, limiter gain-reduction meter, **Mute**, **Ø** (polarity), gain, delay, limiter on/off, threshold and release, and the total latency of that output. Hold ⇧ while changing a value to apply it to all selected strips; ⌘-click names to select several.

EQ editor

Click a strip's name to open its filters under the strips:

- **Graph**: the white curve is the output's total filter response. Drag a numbered handle to move a band (frequency and gain), scroll on it to change Q, double-click an empty spot to add a bell, ⌥-click a handle to delete it. HP / LP / A1… handles move crossovers and all-pass filters.

- **Live TF**: the current measurement, level-matched to the curve.
- **Base** + **predicted**: choose a captured trace; the yellow curve predicts the response after your changes without measuring again.
- **link**: apply edits to all selected strips (e.g. L and R together).
- Panels for PEQ (16 bands), crossover, all-pass (4), air distance, FIR, and speaker presets.

Fault banner

The red *OUTPUTS MUTED* banner appears at start-up, after an interface change or disconnect, and after the audio engine restarts. Nothing plays until you press **Unmute outputs**.

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7 Processing reference

Each output processes its signal in this fixed order:

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fed by (input mix) → gain · polarity → delay → PEQ ×16 → high-pass → low-pass → all-pass ×4 → air shelf → FIR → limiter → mute → interface channel
```

Block	Details	Latency
Gain / polarity / mute	−80…+12 dB; changes ramp over 20 ms (no clicks)	0
Delay	0–1000 ms; changes crossfade over 50 ms	as set
PEQ	16 bands: bell, low/high shelf, notch, low/high-pass 12 dB; 10 Hz–24 kHz, −30…+15 dB, Q 0.1–30	0
Crossover	Butterworth 6/12/18/24/36/48 and Linkwitz-Riley 12/24/48 dB/oct	0
All-pass	Phase only, for alignment: *1st* (180° turn, like a 6 dB/oct crossover), *2nd* (360°, like 12 dB/oct, Q sets how sharply it turns), *LR24-matched* / *LR48-matched* (exactly the phase of that crossover, e.g. to give a full-range box the phase of the sub/top crossover it doesn't have)	0
Air shelf	Automatic high shelf from the air conditions (see below)	0
FIR	WAV or text coefficients (up to 65 536 taps); on/off crossfades	the filter's own (shown)
Limiter	Peak limiter, threshold −40…0 dBFS, no look-ahead	0

IIR filters add no latency; a linear-phase FIR adds half its length (shown on the strip). Filter changes ramp smoothly, so you can adjust EQ during a show.

Air-absorption compensation (AIR)

Air absorbs high frequencies, more over distance and depending on temperature and humidity (ISO 9613-1). Enter the temperature and humidity in the toolbar, press **Tuned now** when you've tuned the system, and give each output its **distance** to its audience (EQ editor). When the air changes, VenueTuner adds a high shelf that restores what the air now takes away (up to +9 dB).

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8 Measurement window ⌘1

Plots

Magnitude: level difference between what the mic hears and the reference, per frequency. Scroll to move the range, double-click to reset.
Phase: timing per frequency, with the arrival time removed. Plotted every 1/48 octave (479 points), because phase turns fast in the highs: 0.1 ms of error is 36° at 1 kHz but 360° at 10 kHz. *Phase view* zooms to ±90° or ±45°; then the wheel moves it and a double-click resets.

Coherence: how much of what the mic hears comes from the system (1 = all). Points below the blanking threshold (*Coh. blank*, default 0.4) are hidden, and fade in up to 0.4 above it. Raise it outdoors or in noise, lower it to see through a reverberant hall.
Impulse response: the arrival in time; the dashed line marks the detected delay.

Measure tab

- **Sound level**, **Mic gain vs reference**: see section 10.
- **View**: *Transfer fn* or *RTA* (real-time spectrum of the mic in 1/3 or 1/6 octave; works on music, no generator needed).
- **Generator**: pink noise on the selected outputs; level in dBFS RMS (maximum –12).
- **Mic / Reference**: the main measurement mic; the reference is the internal loopback (the generator signal, interface latency compensated) or a physical input.
- **Average** (frames), **Smoothing** (magnitude, none to 1/3 octave), **Phase sm.** (phase, separate, default 1/48 octave: keep it fine while the magnitude is smoothed for EQ), **FFT**: *multi-time-window* (long windows in the lows, short in the highs, which hides late reflections) or *fixed 16k*.
- **Log sweep** (`S`): a 1–8 s sweep for a high-resolution measurement and the room's reverberation. It uses the same frequency points and time windows as the live measurement, so a sweep and pink noise of the same speaker read the same; its coherence comes from the signal-to-noise ratio. If a limiter worked during the sweep you are told to lower the level and sweep again.

- **Delay finder**: arrival time in ms and metres. *Store as reference arrival*, then measure another speaker and *Align to ref*. **Lock delay** holds the measurement delay so you can set it by hand: the −0.1 / −0.01 / +0.01 / +0.1 ms buttons move it (nudges under half a sample only rotate the phase; larger ones restart the averaging) and the panel shows what the finder still sees. The usual reason: the impulse peak of a sub or of a filtered box is not where its phase is flat, so nudge until the phase is flat through the crossover region. *Unlock* returns to automatic.
- **Traces**: **Capture** (`C`), show/hide, rename (double-click), average, **Import...** (REW, Smaart or VenueTuner text exports), **Export**.

Matrix tab: sequenced measurements

Groups are outputs played together (e.g. "Subs" = outputs 3 + 4); *One per output* creates them for you. **Positions** are where the mic goes.

- **One mic**: place it, press ▶ in that position's column. Each group plays alone and is captured once its arrival is stable and at least 8 frames are averaged (about 3 s). A capture with low coherence (talking, a door) is retried once, then marked ✕. A capture during which a limiter worked is rejected: lower the generator level. Automatic captures always average at least 16 frames, whatever the display setting. Move the mic and repeat.
- **Several mics**: give each position its own mic and press ▶ **All positions with a mic**: every position is captured at once from the same noise (up to 8 mics).
- Click a cell to re-measure it. **avg** averages a group over all positions and shows it alone, ready for Auto-EQ. Positions are level-matched first, so the one nearest the speaker doesn't outweigh the others.

Room tab

After a log sweep: EDT, T20 and T30 per octave (63 Hz–8 kHz) and the decay curves. Use a quiet room and a 3 s sweep or longer.

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9 Automatic tuning

All automatic tools write **normal settings** you can see and edit, and save a **"Before..." snapshot** first, so you can always go back.

Auto-EQ

1. Measure the output alone (generator on that output only) at one or more positions, and keep those traces visible (or use **avg** in the Matrix).
2. Auto-EQ tab: choose the output, the target (flat, tilt, or house curve with LF lift and HF roll-off), the range (it follows the crossover) and the number of bands.

3 **Compute**: grey = now, yellow = predicted. **Apply** replaces that output's PEQ.

Rules: it prefers cuts; the combined boost never exceeds +3 dB (adjustable); narrow, deep dips are left alone (they are cancellations that EQ can't fix); low-coherence regions are ignored.

Auto delay alignment

Align tab: choose the outputs and the reference (latest arrival or one output). Each output plays alone; an arrival counts once three successive measurements agree within 0.05 ms and the mic clearly hears that speaker. Delays are set so they all arrive together. For tops, fills and delay speakers: subs can't be mixed with full-range outputs here (a sub's impulse peak comes late), use Sub ↔ top below.

Sub ↔ top phase alignment

Capture the top alone and the sub alone from the same mic position. Compute searches the delay **and polarity** that give the strongest sum around the crossover (using phase, not just the impulse peak; low-coherence points count less) and shows the summed response before and after.

The manual way (as done on tour)

1. **Delays first, EQ last.** Time-align, then decide on polarity and all-pass, and only then EQ the sum: EQ on a misaligned system corrects the cancellations instead of the speakers.
2. Measure the main (top) at the alignment position with the delay finder; press **Lock delay** so this time becomes the reference for every following measurement.
3. Mute the top, play the sub alone, keep the locked delay. Capture both traces and compare their phase in the crossover region (Phase sm. 1/24 or finer, Phase view ±90°).
4. Add delay to the earlier of the two (output delay in the EQ editor) until the two phase traces overlap through the crossover. If they are 180° apart, try the polarity first: the "right" delay with the wrong polarity is usually a full period off.
5. Check the sum: the combined magnitude should rise about +6 dB over each part at the crossover. If they still drift apart above or below, an all-pass (often LR-matched on the top) can bend one phase curve onto the other.
6. Repeat the auto estimate a few times: if the finder's value jumps around, there is a reflection or noise problem. Fix that before trusting any delay.

10 Sound level, calibration and limits

Readings

Setting	Options
Frequency weighting	A (hearing, legal limits), C (includes low end, legal limits), Z (flat)
Time weighting	Fast 125 ms, Slow 1 s, Impulse 35 ms
Refresh	1, 2, 5 or 10 times per second (display only; the measurement itself doesn't change)
Leq	Equivalent level over 1 s, 10 s, 1 min, 5 min, 15 min, or since reset (↺)
Also shown	LCpeak, maximum Fast level, time since reset, and the mic level in dBFS

Calibrating

Until a mic is calibrated, levels are an **estimate** (shown with ≈, about ±10 dB), assuming a typical mic and gain. **Calibrate…** offers three methods:

Method	How	Accuracy
1 · Calibrator	Put a 94 dB (or 114 dB) calibrator on the mic, press Calibrate	Best
2 · Reference meter	Play steady pink noise for 10 s with both mics side by side; type the other meter's 10 s Leq (or average Slow reading, its weighting), press Match	As good as the reference
3 · Datasheets	Mic sensitivity (mV/Pa), interface level at 0 dBFS (dBu), preamp gain	Estimate, ±3 dB

> **Calibration follows the gain knob**
> A calibration is valid only at the mic preamp gain it was made at. If you turn the gain, calibrate again. Calibrations are stored per mic name.

Mic gain vs reference

The REF and MIC meters compare the reference (generator / loopback) and the mic. Adjust the interface's mic preamp until the hint shows ✓ *matched* (within about 6 dB). This gives clean, well-scaled transfer functions.

Limits (France, décret 2017-1244)

LAeq over 15 minutes ≤ **102 dB(A)** and LCeq over 15 minutes ≤ **118 dB(C)**; for events aimed at children under 6: 94 dB(A) / 104 dB(C). Click the limits in the Live toolbar to switch preset, enter custom values or turn them off. VenueTuner **displays** the levels; it doesn't limit the audio. Use the output limiters and your own judgement to stay within the law.

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11 Snapshots, presets and undo

	Contains	Use
Snapshot	Inputs, outputs, channels, routing and all processing	A whole show or venue. **Recall** crossfades; **A/B** toggles two snapshots; export/import as files. On another interface you are asked which channels to use.
Speaker preset	Crossover, PEQ, all-pass, limiter, polarity, level and FIR (embedded) of one output	A type of speaker. **Apply** to outputs; delay, routing and channel stay. Export/import `.vtpreset` files to share.
Undo	Every change to inputs, outputs and processing (last 100)	⌘ Z / ⇧ ⌘ Z; a drag counts as one step. The generator and measurement settings are not undone.

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12 Safety and live use

- Outputs start **muted** after launch, after an interface change or disconnect, and after an engine restart.
- **MUTE ALL** acts within one audio block.
- The generator starts at −40 dBFS, fades in and out, and can't exceed −12 dBFS RMS. After a restart it is always off.
- Measurement mics are never routed to outputs.
- Gain, polarity, EQ and delay changes ramp or crossfade; FIR on/off crossfades. Adding, removing or reordering outputs during a show can still glitch: do patch changes before the show.
- If the user interface closes or crashes, the audio keeps running; the app restarts the engine if it stops (outputs muted).
- Use a dedicated Mac for shows. Other audio drivers (virtual soundcards, network audio drivers) can stall macOS audio for every application when they misbehave.

> **Protect the speakers**
> Set limiter thresholds and crossovers before raising amplifier levels. Pink noise and sweeps can be loud: start low and wear hearing protection near the speakers.

13 Keyboard shortcuts

Keys	Action
⌘ 1 · ⌘ 2 · ⌘ ,	Measurement · Live · Settings
⌘ Z · ⇧ ⌘ Z	Undo · redo
G · C · R · S	(Measurement) generator on/off · capture trace · reset averaging · log sweep
⇧ + value change	(Live) apply to all selected strips
Drag · scroll · double-click · ⌥-click	(EQ graph) move band · Q · add band · delete band

14 Troubleshooting

Symptom	Check
"○ NO AUDIO" in the status bar	The interface isn't running: reconnect it, pick it again in Settings › Audio, or quit other apps holding it. If several apps stall at once, restart virtual / network soundcard drivers or the Mac.
Nothing comes out	Red banner → **Unmute outputs**; MUTE ALL; the output's Mute; the output's interface channel; **fed by** in Settings › Outputs; the input type is *program*.
"No clear arrival" / no measurement	Generator on and on the right output; the mic selected and patched; phantom power; mic gain (see *Mic gain vs reference*).
Low coherence, jagged traces	Too much noise in the room or the level is too low: raise the generator, average more, use the multi-time-window FFT, or move the mic away from reflective surfaces.
Clicks or dropouts (XRUN rising)	Use a larger buffer (128 or 256), close other apps, bypass FIR filters on unused outputs.
High latency	Settings › Audio › *Where the latency comes from*: if most of it is "stream (hardware DSP)", it's the device itself (e.g. built-in Mac devices); use a pro interface.
"Outputs on channels this interface doesn't have"	The interface changed: re-patch those outputs/inputs in Settings.
SPL looks wrong	Calibrate (section 10); recalibrate after changing the mic gain. The Leq restarts when you pick another mic. Automatic measurements never touch it.
"Input and output are separate devices" in the status bar	Two interfaces on different clocks drift apart, so internal-loopback delay and phase wander. Use one interface (or an aggregate device with drift correction), or a cabled loopback reference.
macOS refuses to open the app	See section 3 (Open Anyway).

15 Files, data and glossary

Settings, snapshots, speaker presets and calibrations are stored per Mac in `~/Library/Application Support/VenueTuner`. They are saved automatically (every change, and on quit). Snapshots (`.json`) and presets (`.vtpreset`) can be exported and shared.

Term	Meaning
Transfer function (TF)	What the system does to the signal: the ratio of the mic signal to the reference, per frequency (magnitude and phase).
Coherence	0–1: how much of the mic signal is explained by the reference. Low = noise, reflections or too little level.
Internal loopback	The generator signal used directly as the reference, inside the software; the interface latency is compensated.
Linkwitz-Riley (LR)	Crossover filters whose low and high parts sum flat when aligned.
All-pass	A filter that changes phase without changing level; used to align a sub and a top at the crossover.
FIR	A filter defined by its impulse response (coefficients): can correct phase and magnitude independently; may add latency.
LAeq / LCeq	Equivalent continuous sound level, A- or C-weighted, over a period (e.g. 15 min).
RT60 (T20, T30, EDT)	Time for sound to decay by 60 dB in the room, estimated from part of the decay.
dBFS / dB SPL	Level relative to the interface's maximum / acoustic level in the room (needs calibration).

VenueTuner 0.1 · This manual describes the software-only version for macOS.